A project report on

SMART RETAIL ANALYTICS SYSTEM

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology

by

NIYATI GARG (23BCE7071)

RITHIGA M K (23BCE8640)

DATLA MAHEEJA (23BCE8749)

A TARIK (23BCE8799)

School of Computer Science and Engineering

April, 2026

DECLARATION

I hereby declare that the thesis entitled "SMART RETAIL ANALYTICS SYSTEM" submitted by me, for the award of the degree of Bachelor of Technology, VIT-AP University, is a record of bonafide work carried out by me under the supervision of Dr. N Lakshmipathi Anantha.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Amaravati

Date: 30/04/2026 **Signature of the Candidate**

CERTIFICATE

This is to certify that the Senior Design Project titled "**SMART RETAIL ANALYTICS SYSTEM**" that is being submitted by **NIYATI GARG (23BCE7071), RITHIGA M K (23BCE8640), DATLA MAHEEJA (23BCE8749), and A TARIK (23BCE8799)** is in partial fulfillment of the requirements for the award of Bachelor of Technology, is a record of bonafide work done under my guidance. The contents of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for award of any degree or diploma and the same is certified.

Dr. N Lakshmipathi Anantha
Internal Guide

The thesis is satisfactory / unsatisfactory

Internal Examiner **External Examiner**

Approved by

DEAN
School Of Computer Science and Engineering

ACKNOWLEDGEMENT

It is my pleasure to express with deep sense of gratitude to Dr. N Lakshmipathi Anantha, Associate Professor Grade 1, SCOPE, VIT-AP, for his constant guidance, continual encouragement, understanding; more than all, he taught me patience in my endeavor. My association with him is not confined to academics only, but it is a great opportunity on my part of work with an intellectual and expert in the field of Artificial Intelligence.

I would like to express my gratitude to G. Viswanathan, Dr. Sekar Viswanathan, Dr. Sankar Viswanathan, Dr. G. V. Selvam, Dr. P.Arulmozhivarman, and Dr. S.Sudhakar Ilango, SCOPE, for providing with an environment to work in and for his inspiration during the tenure of the course.

In jubilant mood I express ingeniously my whole-hearted thanks to Dr. G.Muneeswari, HOD, Department of Data Science and Engineering, all teaching staff and members working as limbs of our university for their not-self-centered enthusiasm coupled with timely encouragements showered on me with zeal, which prompted the acquirement of the requisite knowledge to finalize my course study successfully. I would like to thank my parents for their support.

It is indeed a pleasure to thank my friends who persuaded and encouraged me to take up and complete this task. At last but not least, I express my gratitude and appreciation to all those who have helped me directly or indirectly toward the successful completion of this project.

Place: Amaravati

Date: **Name of the student**

CONTENTS

TABLE OF CONTENTS

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

With the rapid advancement of computer vision and artificial intelligence, traditional surveillance systems are evolving into intelligent analytics platforms. In retail environments, understanding customer behavior is crucial for improving store layout, optimizing product placement, and enhancing overall customer experience. However, most retail stores still rely on conventional CCTV systems that only provide video footage without meaningful insights.

Modern technologies such as deep learning-based object detection and tracking enable automated analysis of video data. These technologies allow businesses to move beyond passive monitoring and towards data-driven decision-making.

1.2 OVERVIEW OF SMART RETAIL ANALYTICS

Smart retail analytics refers to the use of advanced computational techniques to extract actionable insights from retail data, particularly video streams. By integrating computer vision models with analytics tools, it is possible to monitor customer activity, track movement patterns, and generate useful metrics such as footfall, dwell time, and zone popularity.

In this project, a Smart Retail Analytics System is developed that processes CCTV video streams to automatically detect, track, and analyze customer behavior in real time.

1.3 PROBLEM STATEMENT

Traditional retail monitoring systems rely on manual observation or basic counting mechanisms such as entry-point sensors. These approaches have several limitations:

- They cannot track customer movement within the store.
- They fail to calculate zone-wise dwell time.
- They provide inaccurate counts in crowded environments.
- They do not support real-time analytics or decision-making.

As a result, valuable insights about customer behavior remain unavailable, leading to inefficient store management and missed opportunities for optimization.

1.4 OBJECTIVES OF THE PROJECT

The primary objective of this project is to develop an automated system that utilizes computer vision techniques to analyze customer behavior in retail environments. The system aims to:

- Detect customers in video streams using deep learning models.
- Track individuals across frames while maintaining unique identities.
- Perform zone-based analysis to study customer movement patterns.
- Calculate metrics such as footfall, dwell time, and zone popularity.
- Provide real-time and offline analytics for better decision-making.

1.5 SCOPE OF THE PROJECT

The system is designed to operate on CCTV video feeds in both offline and live modes. It focuses on analyzing customer behavior within a defined retail space by dividing the area into zones and tracking movement across these zones. The scope includes multi-camera video processing, customer tracking and identification, and behavioral analysis using zone mapping. However, the system does not focus on facial recognition, ensuring privacy concerns are mitigated.

1.6 SYSTEM OVERVIEW

The proposed system follows a structured pipeline for processing video data:

1. **Detection:** Customers are detected in video frames using YOLOv8.
2. **Tracking:** Detected individuals are tracked across frames using tracking algorithms.
3. **Re-Identification:** Unique identities are maintained to avoid duplicate counting.
4. **Zone Analysis:** The store is divided into grids to analyze movement and dwell time.
5. **Analytics Generation:** Insights such as footfall and zone popularity are generated.

1.7 SIGNIFICANCE OF THE PROJECT

The proposed system eliminates the need for manual observation, provides accurate real-time analytics, and helps optimize store layout. By transforming raw video data into meaningful insights, the system enables smarter, data-driven retail management.

CHAPTER 2
LITERATURE SURVEY

2.1 INTRODUCTION

With the advancement of computer vision and deep learning, automated video analytics has become a key area of research in retail environments. Traditional surveillance systems generate large volumes of video data, but extracting meaningful insights requires intelligent techniques such as object detection, tracking, and person re-identification. This chapter reviews existing research work relevant to these components.

2.2 OBJECT-DETECTION USING YOLO

Object detection plays a crucial role in identifying customers in video streams. The work by *You Only Look Once: Unified, Real-Time Object Detection* (Redmon et al., 2016) introduced a unified approach that performs detection in a single forward pass, making it highly efficient for real-time applications. YOLO-based models are widely adopted in surveillance and retail analytics due to their speed and accuracy. However, these models are limited to detecting objects and do not provide tracking or identity persistence.

2.3 MULTI-OBJECT TRACKING USING DEEPSORT

To address the limitation of detection-only systems, tracking algorithms such as *Simple Online and Realtime Tracking with a Deep Association Metric* (Wojke et al., 2017) have been proposed. DeepSort enhances tracking by assigning unique IDs to detected individuals and maintaining identity across frames using motion and appearance features. While effective in single-camera environments, it does not support identification of the same individual across multiple cameras.

2.4 PERSON RE-IDENTIFICATION (REID)

Person Re-Identification aims to recognize the same individual across different camera views. Studies such as *Person Re-identification: Past, Present and Future* (Zheng et al., 2016) highlight the use of deep learning models to generate feature embeddings that uniquely represent individuals. These embeddings are compared using similarity measures to identify matches. Despite advancements, challenges such as variation in lighting, pose, and occlusion affect accuracy.

2.5 RETAIL ANALYTICS AND CUSTOMER BEHAVIOR ANALYSIS

Research in retail analytics focuses on understanding customer behavior through video data. Early work such as *Video Analytics for Retail* (Senior et al., 2007) explored automated analysis of customer activities in retail environments. Later studies such as *Customer Behavior Recognition in Retail Stores from Surveillance Cameras* (Liu et al., 2015) and *Deep Understanding of Shopper Behaviours and Interactions Using RGB-D Vision* (Paolanti et al., 2020) introduced methods for analyzing movement patterns, dwell time, and interactions within stores.

More recent approaches, such as *Advanced Customer Behavior Tracking and Heatmap Analysis with YOLOv5 and DeepSORT in a Retail Environment* (Shili et al., 2024), combine object detection and tracking to generate heatmaps and behavioral insights. However, these systems are often limited to single-camera setups and lack cross-camera identity matching.

2.6 RESEARCH GAP

From the above literature, it is evident that most systems focus on individual components such as detection, tracking, or behavior analysis. However, there is a lack of integrated systems that combine:

- Real-time object detection and tracking
- Cross-camera identity matching using ReID
- Customer behavior analytics (zone analysis, dwell time)
- Visualization and reporting of insights

Additionally, many existing systems fail to provide a unified pipeline that converts raw video data into actionable business insights.

CHAPTER 3
EXISTING WORK

3.1 INTRODUCTION

Retail stores have traditionally relied on surveillance systems for security and monitoring purposes. These systems primarily consist of CCTV cameras that continuously record video footage. While useful for observation, they lack the capability to automatically analyze and extract meaningful insights from the collected data.

3.2 TRADITIONAL CCTV-BASED SYSTEMS

Conventional retail monitoring systems use CCTV cameras to capture video data, which is manually reviewed by store personnel when required. These systems provide basic functionalities such as live monitoring and recording but do not support automated data analysis.

The process typically involves:

- Continuous video recording
- Manual observation or playback
- Limited incident-based review

This approach is time-consuming and inefficient, especially in large retail environments.

3.3 LIMITATIONS OF EXISTING SYSTEMS

The existing systems suffer from several major limitations:

1. Lack of Automation

Traditional systems require human intervention to analyze video footage, making the process inefficient and prone to errors.

2. No Customer Behavior Insights

These systems do not provide information such as:

- Customer movement patterns
- Time spent in specific areas
- Popular zones within the store

3. No Identity Tracking

Conventional systems cannot track individual customers across frames or cameras. Even advanced systems using tracking algorithms are limited to single-camera environments.

4. Absence of Cross-Camera Matching

Existing solutions fail to identify the same person across different cameras, which is essential for large retail stores.

5. Lack of Data Visualization

There is no structured output such as charts, reports, or dashboards to help store managers make data-driven decisions.

6. Inefficient Use of Data

Large amounts of video data are generated but remain underutilized due to the lack of analytical tools.

3.4 NEED FOR IMPROVEMENT

The limitations of existing systems highlight the need for an intelligent solution that can automatically process video data and generate actionable insights. Modern retail environments require systems that can:

- Detect and track customers in real time
- Analyze customer behavior
- Identify individuals across multiple cameras
- Provide visual insights through dashboards and reports

3.5 CONCLUSION

Thus, existing retail monitoring systems are insufficient for advanced analytics and decision-making. There is a clear need for a smart retail analytics system that overcomes these limitations by integrating computer vision and machine learning techniques.

CHAPTER 4
PROPOSED WORK

4.1 INTRODUCTION

The proposed work of this project is to develop an intelligent Customer Analytics System (CAP) using CCTV surveillance, Computer Vision, and Deep Learning techniques to analyze customer behavior inside retail stores. The system is designed to provide accurate visitor counting, customer movement tracking, zone-based heatmap generation, and real-time monitoring to support better business decision-making.

4.2 SYSTEM OVERVIEW

The project utilizes multiple CCTV camera feeds installed inside the store and processes them using YOLOv8 for human detection, DeepSORT for object tracking, and Re-Identification (ReID) models to recognize the same customer across different cameras. This helps prevent duplicate counting and improves the accuracy of visitor analysis.

The system follows a pipeline-based approach where video input is processed step-by-step to extract meaningful insights.

4.3 ZONE-BASED ANALYSIS

To understand customer movement patterns, the store layout is divided into an 8×4 zone grid system. The system tracks customer movement across these zones and identifies high-traffic and low-traffic areas, frequently visited sections, and customer dwell time in each zone. Based on this analysis, heatmaps are generated to visually represent customer activity and shopping behavior.

4.4 CUSTOMER SESSION TRACKING

The project also includes session tracking for each visitor, which records entry time, exit time, total time spent inside the store, and movement history. This helps in understanding customer engagement and shopping patterns more effectively.

4.5 ADVANCED FEATURES AND SCALABILITY

As a future enhancement, the project includes LIVETRACK, a real-time customer monitoring system using RTSP-enabled Hikvision cameras for 24/7 live tracking, crowd monitoring, and instant alerts. The system also supports MUTISTORE architecture, allowing multiple store branches to be monitored from a centralized dashboard.

4.6 OBJECTIVES OF PROPOSED SYSTEM

The overall goal of the proposed work is to convert traditional CCTV systems into intelligent analytics tools that help businesses:

- Optimize store layout
- Improve customer experience
- Enhance operational efficiency
- Increase overall sales performance

The overall goal of the proposed work is to convert traditional CCTV systems into intelligent analytics tools that help businesses optimize store layout, improve customer experience, enhance operational efficiency, and increase overall sales performance.

4.7 ADVANTAGES OF PROPOSED SYSTEM

Compared to existing systems, the proposed system offers:

- Automated customer detection and tracking
- Cross-camera identity matching using ReID
- Accurate visitor counting without duplication
- Detailed zone-wise analytics and heatmaps
- Scalable architecture for multiple stores

4.8 CONCLUSION

Thus, the proposed system provides a comprehensive and intelligent solution for retail analytics by integrating detection, tracking, and behavioral analysis into a unified framework.

CHAPTER 5
IMPLEMENTATION

5.1 INTRODUCTION

The implementation of this project is based on using the existing CCTV cameras inside retail stores to study and understand customer behavior. The system continuously collects video from different cameras placed in various sections of the store and uses that footage to track customer movement and shopping patterns.

5.2 DATA ACQUISITION

The system captures video input from multiple CCTV cameras installed across the store. These cameras provide continuous video streams, which serve as the primary data source for analysis.

5.3 CUSTOMER DETECTION AND TRACKING

The first step is detecting the customers present in each video frame. Once a person is detected, the system follows their movement from one frame to another and assigns a unique identity so that the same person can be tracked throughout their visit. This helps in maintaining proper customer counting and prevents confusion when multiple people are present at the same time.

5.4 CROSS-CAMERA IDENTIFICATION

Since customers may move between different camera views, the system also checks whether the same person appears in another camera. This avoids counting the same customer more than once and improves the overall accuracy of visitor analysis.

5.5 ZONE-BASED MOVEMENT ANALYSIS

To study customer movement in detail, the store is divided into small sections using a fixed grid layout. Whenever a customer moves through the store, the system records which sections they visit and how much time they spend there. This helps identify which areas attract more customers and which areas receive less attention.

5.6 CUSTOMER SESSION RECORDING

The system also records the complete visit of each customer, including when they enter the store, when they leave, how long they stay, and the path they follow inside the store. This information helps in understanding customer engagement and shopping behavior more clearly.

5.7 DATA STORAGE AND PROCESSING

All the collected information is stored in structured formats such as CSV files and processed further to generate analytics. These datasets include visitor details, zone-wise time, and movement paths.

5.8 VISUALIZATION AND DASHBOARD

The processed data is displayed through a dashboard where store owners can view visitor counts, movement reports, and area-based activity. This makes it easier to improve product placement, store layout, and overall customer experience.

5.9 CODE OVERVIEW

This section presents key code snippets used in the implementation of the Smart Retail Analytics System. Instead of including the entire codebase, only important modules and logic are highlighted along with descriptions for clarity.

1. Video Upload and Processing (Flask Backend)

Figure 5.1: Video Upload Handling in Flask

```python
@app.route('/upload', methods=['POST'])
def upload():
    if 'video' not in request.files:
        return "No video files uploaded", 400

    files = request.files.getlist('video')
    saved_paths = []

    for file in files:
        if file and allowed_file(file.filename):
            filename = secure_filename(file.filename)
            save_path = os.path.join(app.config['UPLOAD_FOLDER'], filename)
            file.save(save_path)
            saved_paths.append(save_path)

    run_customer_analysis_multi(saved_paths, output_folder)
```

Description:

This code handles video uploads from the user interface. Multiple video files are accepted, saved securely, and passed to the main analysis function for further processing.

2. Customer Detection using YOLO

Figure 5.2: YOLO-based Person Detection

```
results = yolo(frame, verbose=False)
detections = []
for result in results:
    for box in result.boxes:
        if int(box.cls[0]) == 0:
            x1, y1, x2, y2 = map(int, box.xyxy[0])
            detections.append(([x1, y1, x2 - x1, y2 - y1], 0.99, "person"))
```

Description:

This snippet uses the YOLO model to detect objects in each video frame. Only detections classified as "person" are selected and stored for tracking.

3. Multi-Object Tracking using DeepSort

Figure 5.3: DeepSort Tracking

```
tracks = []
if detections:
    tracks = tracker.update_tracks(detections, frame=frame)

for track in tracks:
    if not track.is_confirmed():
        continue
    tid = f"{cam_id}_{track.track_id}"
```

Description:

DeepSort assigns unique IDs to each detected person and tracks them across frames. This ensures that each customer is consistently identified during their movement.

4. Zone Mapping and Movement Tracking

Figure 5.4: Zone Calculation Logic

```
cx = int((l + r) / 2)
cy = int((t + b) / 2)
if cx < 0 or cy < 0 or cx >= width or cy >= height:
    continue
z_row = min(max(cy // zone_h, 0), ZONE_ROWS - 1)
z_col = min(max(cx // zone_w, 0), ZONE_COLS - 1)
zone_key = f"Zone_{z_row}_{z_col}"
```

Description:

The store layout is divided into a grid. This code calculates the zone in which a customer is located based on their position in the frame.

5. Feature Extraction for Re-Identification

Figure 5.5: ReID Feature Extraction

```
def extract_reid_feature(image):
    if not is_valid_crop(image):
        return None
    img = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    img = transform(img).unsqueeze(0).to("cpu")
    with torch.no_grad():
        feature = reid_model(img)
    return feature.squeeze(0).cpu().numpy()
```

Description:

This function extracts feature vectors from customer images using a deep learning model. These features are later used to identify the same person across different cameras.

6. Cross-Camera Matching using Cosine Similarity

Figure 5.6: Similarity Calculation

```
vectors = np.array([v["feature"] for v in feature_data])
vectors = normalize(vectors)
similarity = cosine_similarity(vectors)
```

Description:

This code computes similarity between feature vectors using cosine similarity. Higher similarity indicates that two detections likely belong to the same person.

7. Visitor Merging Logic

Figure 5.7: Grouping Matched Visitors

```
grouped_data = {}
for i, m_id in enumerate(matched):
    grouped_data.setdefault(m_id, []).append(feature_data[i])
```

Description:

Visitors identified as the same person are grouped together using matching IDs. This helps in merging duplicate entries across cameras.

8. Heatmap Generation

Figure 5.8: Heatmap Creation

```
zone_df = pd.DataFrame(zone_heatmap.astype(int))
plt.figure(figsize=(10, 6))
sns.heatmap(zone_df, cmap="YlOrRd", annot=True, linewidths=0.5, cbar=True)
plt.title("Zone-based Customer Heatmap")
plt.xlabel("Zone Columns")
plt.ylabel("Zone Rows")
plt.savefig(os.path.join(output_folder, "zone_heatmap.jpg"))
```

Description:

This code generates a visual heatmap showing customer activity across different zones of the store.

9. Dashboard Visualization

Figure 5.9: Chart Generation (Frontend)

```
if (visitorChart) visitorChart.destroy();
visitorChart = new Chart(document.getElementById("visitorTimeChart"), {
  type: "bar",
  data: {
    labels: visitorLabels,
    datasets: [{ label: "Time (s)", data: visitorDurations, backgroundColor: "rgba(54, 162, 235, 0.6)" }]
  },
  options: { responsive: true, plugins: { legend: { display: false } } }
});
```

```
if (zoneChart) zoneChart.destroy();
zoneChart = new Chart(document.getElementById("zoneTimeChart"), {
  type: "bar",
  data: {
    labels: Object.keys(zoneTotals),
    datasets: [{ label: "Total Time (s)", data: Object.values(zoneTotals), backgroundColor: "rgba(255, 99, 132, 0.6)" }]
  },
  options: { responsive: true, plugins: { legend: { display: false } } }
});
```

Description:

The dashboard uses charts to display analytics such as time spent by each visitor and zone-wise activity.

5.10 CONCLUSION

The code implementation integrates multiple components including video processing, object detection, tracking, re-identification, and data visualization. By combining these modules, the system successfully converts raw CCTV footage into meaningful customer analytics.

CHAPTER 6
OUTPUT

6.1 INTRODUCTION

The output of the Smart Retail Analytics System is presented through a user-friendly interface and visual representations that help in understanding customer behavior effectively. The system generates multiple outputs at different stages of processing.

6.2 UPLOAD INTERFACE

The system provides an upload page where users can select and upload video files from CCTV cameras. This interface allows multiple video inputs to be processed simultaneously.

6.3 PROCESSING STAGE

Once the videos are uploaded, the system enters a processing stage where customer detection, tracking, and analysis are performed. A loading or processing screen is displayed to indicate that the system is analyzing the data.

6.4 DATA DASHBOARD

After processing, the system displays a dashboard that presents key analytics such as:

- Total number of visitors
- Time spent by each visitor
- Zone-wise activity

The dashboard uses graphical representations such as bar charts for better understanding.

6.5 SUMMARY REPORT

The system generates a summary report that includes:

- Total visitor count
- Average time spent by customers

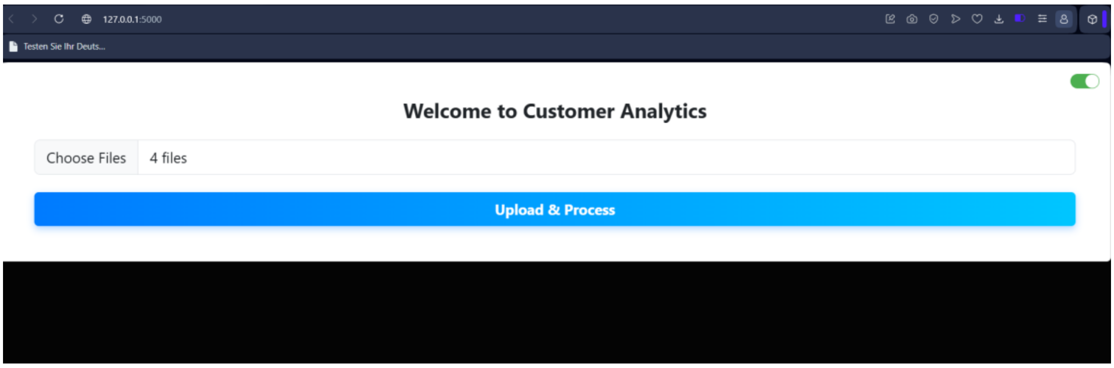
This report provides a quick overview of store activity.

6.6 ZONE-BASED HEATMAP

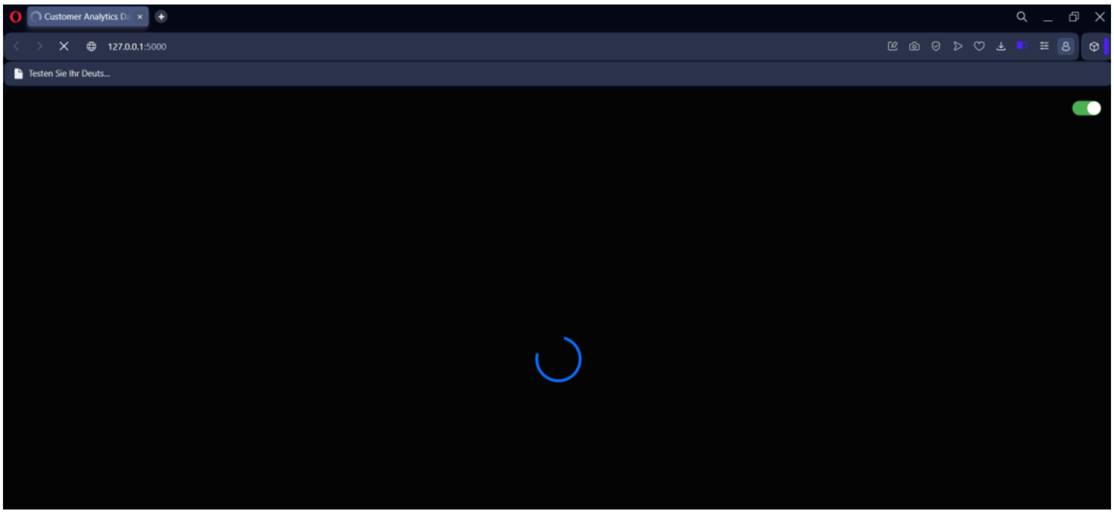
A heatmap is generated to visually represent customer activity across different zones of the store. High-traffic areas are highlighted, helping store owners identify popular sections and optimize layout accordingly.

6.7 RESULTS

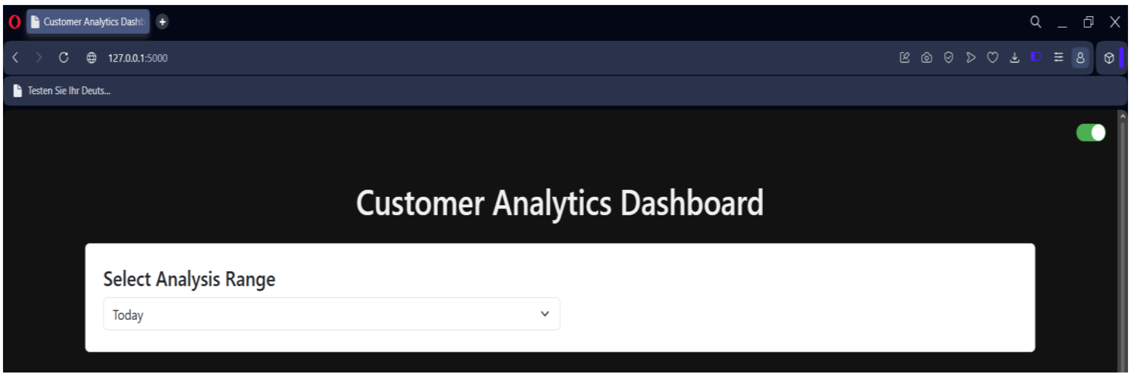
1. Upload Page Interface

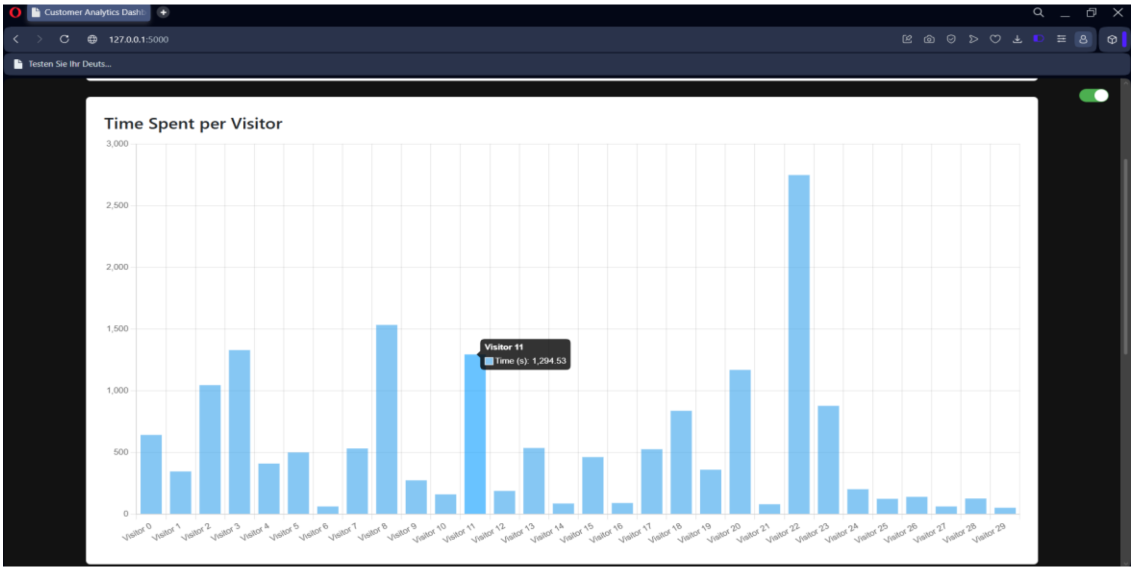


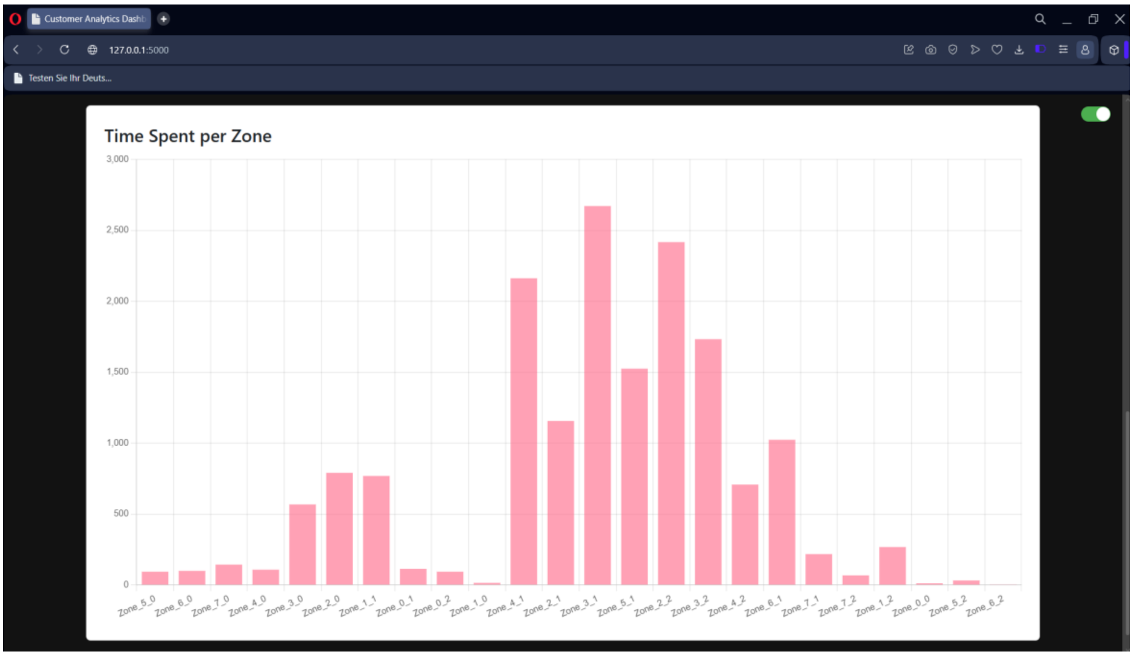
2. Processing Screen



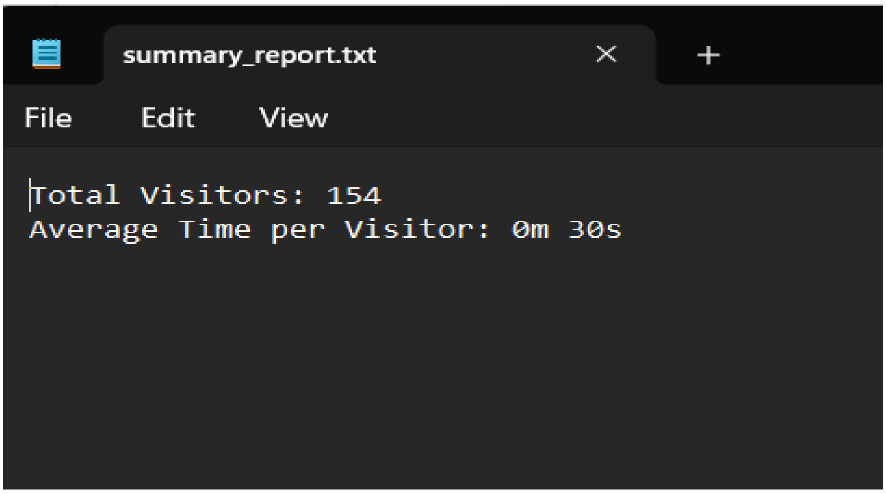
3. Dashboard View







4. Summary Report



5. Zone Based Heatmap

6.8 CONCLUSION

The outputs generated by the Smart Retail Analytics System effectively transform raw CCTV footage into meaningful and actionable insights. Through the dashboard, summary reports, and zone-based heatmaps, the system provides a clear understanding of customer movement patterns, time spent, and area-wise engagement within the store. These visual and data-driven outputs enable store owners to identify high-traffic zones, optimize product placement, and improve overall store layout. The system not only enhances decision-making but also reduces reliance on manual observation by providing accurate and automated analytics.

Overall, the output module demonstrates the practical usefulness of the system in real-world retail environments, making it a valuable tool for improving operational efficiency and customer experience.

CHAPTER 7
REFERENCES

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). *You only look once: Unified, real-time object detection*. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

Wojke, N., Bewley, A., & Paulus, D. (2017). *Simple online and realtime tracking with a deep association metric*. In Proceedings of the IEEE International Conference on Image Processing (ICIP).

Zheng, L., Yang, Y., & Hauptmann, A. G. (2016). *Person re-identification: Past, present and future*. arXiv preprint arXiv:1610.02984.

Senior, A., Brown, L., Hampapur, A., Shu, C. F., Lu, M., Velipasalar, S., Tian, Y. L., & Ekin, A. (2007). *Video analytics for retail*. In Proceedings of the IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS). IEEE.

Liu, J., Gu, Y., & Kamijo, S. (2015). *Customer behavior recognition in retail stores from surveillance cameras*. In Proceedings of the IEEE International Symposium on Multimedia (ISM). IEEE.

Paolanti, M., Pietrini, R., Mancini, A., Frontoni, E., & Zingaretti, P. (2020). *Deep understanding of shopper behaviours and interactions using RGB-D vision*. Machine Vision and Applications, 31(66). Springer.

Obermeier, G., & Auinger, A. (2019). *Human-computer interaction in physical retail environments and the impact on customer experience: Systematic literature review and research agenda*. In HCI in Business, Government and Organizations. Springer.

Shili, M., Jayasingh, S., & Hammedi, S. (2024). *Advanced customer behavior tracking and heatmap analysis with YOLOv5 and DeepSORT in a retail environment*. Electronics, 13(23), 4730.

Dhanushkodi, K., Bala, A., Kodipyaka, N., & Shreyas, V. (2025). *Customer behavior analysis and predictive modeling in supermarket retail: A comprehensive data mining approach*. IEEE Access, 13, 2945–2957.

Ultralytics. (2024). *YOLOv8 documentation*. Retrieved from https://docs.ultralytics.com/

OpenCV. (2024). *OpenCV documentation*. Retrieved from https://docs.opencv.org/

PyTorch. (2024). *PyTorch documentation*. Retrieved from https://pytorch.org/docs/

TensorFlow. (2024). *TensorFlow documentation*. Retrieved from https://www.tensorflow.org/

Hikvision. (2024). *Camera integration and RTSP configuration guide*.